Assessment Task 1 Unit Plan – CPA 2

Introduction
Model Farms High School is a NSW government secondary school located in Baulkham Hills, NSW. This program of work is based in Model Farms' preliminary Industrial Technology multimedia course, which has a wide variety of students with different learning needs and abilities. Students within this course include approximately 45% of students with a background other than English and 1% of students from an Indigenous background (ACARA, 2024), as well as students with intellectual, psychological and physical disabilities As a result, differentiation has been implemented where possible through the unit to enhance student learning, however, due to the nature of the unit being primarily split between 3 software programs, some differentiation has not been possible. This unit of work has been developed using the process of backward mapping, from the NSW Education Standards Authority (NESA) Industrial Technology syllabus (2013), drawn from resources from the Crestwood High School unit: Unit Two: 2D and 3D animation (Henderson, 2024). The aim of this unit is to provide students with the skills and knowledge in animation, both 2D and 3D, as well as provide critical WHS and industry considerations related to animation. The unit draws its basis from Vygotsky's Zone of proximal development through scaffolding (Wass & Golding, 2014), humanism through student-centred learning (Tangney, 2013) and Bloom's taxonomy (Tabrizi & Rideout, 2017) to build a strong basis for effective teaching and learning. Students will participate in a range of theoretical and practical activities that aim to develop their skills in preparation for Unit 3 where they will develop a mini major project, as well as the major project in the Industrial Technology HSC course. Elements of this program are highly scaffolded, and teacher guided to allow student success in skill development, learning and the completion of the summative assessment within this unit, which consists of two animation projects and two portfolios.
Rationale and Justification
This 10-week preliminary unit of work includes various teaching, learning and assessment methods to maximise student development. Student-led learning following teacher demonstration forms a key part of the unit, as students observe, practise and implement learning and skills. This method of learning is highly effective animation, as it allows students to become very familiar with the software they are using and develop their own understanding of how to apply skills (Yuen, et al., 2018). In weeks 2-4, this learning is seen through the demonstration of 2D skills, followed by independent student skill development, building highly relevant practical and theoretical skills useful for the HSC course. This format build towards the unit's summative assessments, as students independently develop skills over the unit to develop a 2D and 3D animation for the summative assessments. The skills displayed in these summative assessments are critical for the improvement of the unit in future years, as the teacher can ascertain the degree of skill that students have developed, as well as a critical learning opportunity for students in preparation for the HSC course. On-going evaluations, as introduced in Week 1, is an effective Assessment for formative assessment strategy, assisting students in evaluating their progression and learning. These evaluations provide the teacher with information on whether students are understanding the content or falling behind (OECD publishing, 2005). On-going evaluations is further an effective method of providing students with a guide to use when developing their evaluations at the end of each summative assessment tasks, resulting in a more realistic and logical final evaluation. Weekly worksheets act as a summary method in Weeks 1-8 to provide further 'Assessment For' formative method, additionally allowing the teacher to understand where student knowledge may be lacking, providing opportunity for improvement during the unit (OECD publishing, 2005). A variety of resources have been utilised within the unit, accommodating for all types of learners within the classroom, whilst also providing students with relevant, real-world applications of their learning. These resources include the double diamond design process and 12 principles of animation (Becker, 2017) which

provide a logical design structure and format that students can follow step-by-step. Worksheets, videos, PowerPoint presentations and hands-on activities are utilised as information presentation techniques to allow for varied learning types. This is explicitly seen in week 9, where students watch a video by Kim Beaton, from the Weta Workshop (Beaton, 2024), on prototyping using aluminium foil. Students watch the demonstration video then attempt to prototype independently, with the option to refer to the video, or written instructions where necessary. The combination of presentation methods in this lesson facilitates engagement and participation from each student in the classroom (Alabi, 2024)

Pedagogical frameworks within this unit form the basis for effective teaching and learning. This unit has been based around Bloom's taxonomy (Tabrizi & Rideout, 2017) and Vygotsky's Zone of Proximal Development (ZPD) (Wass & Golding, 2014) with elements of humanism. Throughout the unit, elements of higher order thinking are induced by active learning methods, particularly creating and evaluating are critical to effective student learning. This is particularly seen through the on-going evaluations, which encourage students to critically think about their skills and progress, and the evaluations within both summative assessments. These activities provide students with an opportunity to reflect upon and evaluate the products that they have created during the unit (Tabrizi & Rideout, 2017). Student creativity is seen and encouraged throughout the entire unit, with students creating their own designs for both summative assessments, which they work on each week within this unit of work. The inclusion of scaffolding throughout the unit, particularly within the mini portfolios included in the summative assessments follow the basis of Vygotsky's ZPD, encouraging students to learn and grow with effective and balanced guidance (Wass & Golding, 2014). Student-led learning is encouraged in accordance with Humanism, increasing student self-efficacy and self-actualisation throughout the unit as students develop and discover their skills, ending the unit with a successful and tangible product (Serra, et al., 2023).

A variety of teaching strategies, including the integration of general capabilities, have bene utilised within this 10-week unit to maximise effective student learning. Techniques including think-pair-share, brainstorming and a gradual release of responsibility are utilised to encourage student engagement and higher order thinking. Think-pair-share, as seen in week two, and brainstorming, seen in week four, allow students to work collaboratively to create, refine and share valuable ideas, encouraging student engagement and deeper understanding (Guenther & Abbott, 2024). The gradual release of responsibility model is utilised within each week, as the teacher demonstrates a new skill, students work through the example with the teacher, before being provided with the opportunity to work independently for the remainder of the lesson. This is highly effective, as students can comprehend a larger amount through the progressive steps and a reduced cognitive load (Lin & Cheng, 2010).

Digital literacy (ICT) is highly relevant within this unit, as the unit is based on digital platforms and software. Throughout the unit, students develop skills in a range of areas of ICT including digital design software, digital WHS considerations, typing abilities and other software applications. ICT is a key learning point within the 21st century and a valuable resource in teaching as it can increase student engagement and accessibility, provide more resources and encourage more critical thinking (Kumar, 2020). The inclusion of online worksheets in weeks 1-8 is highly beneficial as it offers an opportunity for students to develop their typing abilities and basic software skills, as well as providing an easier method of data collection and submission.

Literacy has been incorporated within this unit of work through the mini portfolios in the summative assessments, and the weekly worksheets. The mini portfolios in Assessment 2 both require students to write with a formal paragraph structure, aiming to develop not only their knowledge in Industrial technology, but also their general writing and literacy skills. Paragraph tasks are seen within the Statement of Intent and Evaluation work that students complete in Weeks 2, 5, 6 and 10, and require students to use subject specific language and formal register that they will develop throughout the entire unit.

Numeracy has been particularly utilised in weeks 2 and 5, as students encounter keyframing, frame rates and file sizes. In week 2, students will investigate keyframing and the basic concept of frame rates, which are critical

within animation, as they determine the quality of the animation. Students will develop basic numeracy skills in calculating how many keyframes they will require for a high-quality project. In week 5, students will investigate frame rates, sample rates and file sizes, particularly in relation to audio. They will spend their theory lessons throughout this week completing calculations of increasing difficulty, applying and developing numeracy skills to solve relevant, real-world problems, encouraging greater engagement and connection with the content (Victoria State Government, 2024).
Professional and ethical responsibilities within this unit have been considered through classroom safety, cyber safety, AI considerations, and adjustments for diverse learners. Prior to student's commencing classes, WHS within the classroom has been considered, and as a result, trip hazards and wires have been removed, or secured, as seen in the Work Health and Safety considerations of the unit. Students are asked to consider and develop strategies to manage WHS issues both online and within the classroom in week 5, presenting this information in summative assessment 2.1. Appropriate cyber safety methods including multi-factor authentication and a firewall are additionally in place as organised by the Department of Education NSW to protect students and their movements online. Considerations for the use of AI include the personalised nature of the summative assessment tasks, which removes the ability for students to use AI, as they must develop and document all their processes. Despite this, AI has been incorporated into some elements of the unit, due to its value as a classroom resource, students will learn how to effectively use AI to scaffold their learning, and how to apply the information they receive effectively. Additionally, the completion of all work during class time allows the teacher to monitor students ICT usage. Adjustments within learning have been provided throughout the entire unit to support a range of learners within the classroom. This is evident within the summative assessment tasks, which have both been scaffolded to varying degrees for different levels of learners. This allows students with work within their Zone of Proximal Development (ZPD) (Wass & Golding, 2014), effectively building their skills and knowledge throughout the unit. Adjustments are also seen within the extension task offered in weeks 5 and 10, as well as quick finisher tasks in the weekly worksheets. Extension and enrichment tasks provide high potential students with the ability to challenge themselves and achieve higher learning throughout the unit (Howell, 2020). The extension task provided is a Stop Motion task, which is a multimedia technique that student's may use for their HSC major project, however, is a difficult skill to develop, as such providing high potential students with a challenging, but engaging opportunity for learning. Throughout the unit of work, varied presentation and learning methods are provided to allow for effective learning of each student in the classroom. Students will learn through class discussions, small group work, paired work and independent work, as well as having the opportunity to choose whether they work independently or in pairs through the weekly worksheets. This allows students to work to their full abilities, with some students being suited to independent work, whilst some students benefit from the knowledge a partner can provide (Singh, et al., 2024). Aural, visual, written and hands-on work is also provided throughout the entire unit, such as in the prototyping activity in week 9, to accommodate for varied learning styles in the classroom. The incorporation of a range of diverse learning methods and styles accommodates the wide range of learners within this preliminary class, resulting in highly effective teaching and learning throughout the whole unit of work.

Model Farms High School, Baulkham Hills

Industrial Technology: Multimedia Technologies

Stage 6

Year 11 Preliminary course

<table>
<tr><td colspan="2">Overview
Unit Length: 10 Weeks (Term 2, Weeks 1-10)
Digital Media has become increasingly popular in the 21st century, with many jobs moving towards computer-based applications.

This unit will provide students with the skills and knowledge in animation, covering 2D and 3D animation techniques, software, hardware and the incorporation of audio. They will additionally learn about the importance of research and planning within project develop and such applications in industry settings. This unit will provide them with the basic skills and knowledge that they will require for their HSC course, real-world applications and any future tertiary studies. The unit assessments are guided by course outcomes and descriptors and demonstrate both theoretical and practical skills that students will develop within this unit of work.</td></tr>
<tr><td>Assessment:

Formal:
Assessment Task 2: Animation Projects and Portfolios (20%)
Assessment Task 2.1: Mini 2D animation project and portfolio (10%) – Weeks 1 -5, Due last lesson of Week 5
Assessment Task 2.2: Mini 3D animation project and portfolio (10%) – Weeks 6-10, due last lesson of Week 10
Unit content will additionally be assessed in Assessment Task 3: Mini Major Project in Term 3 – 30%
Informal:
Weekly self-evaluation
Participation and understanding of weekly practical activities
Weekly worksheets</td><td>Preliminary Course Outcomes

P1.2 Identified appropriate equipment, production and manufacturing techniques, including new and developing technologies
P2.1 Describes and uses safe working practices and correct workshop equipment maintenance techniques
P3.1 Sketches, produces and interprets drawings in the production of projects
P3.2 Applies research and problem-solving skills
P3.3 Demonstrates appropriate design principles in the production of projects
P4.1 Demonstrates a range of practical skills in the production of projects
P4.2 Demonstrates competency in using relevant equipment, machinery and processes
P5.2 Uses appropriate documentation techniques related to the management of projects
P6.1 Identifies the characteristics of quality manufactured products</td></tr>
<tr><td colspan="2">Work Health and Safety Considerations
Students are expected to follow safe ICT behaviours whilst using computers during these lessons. They will be required to consider and practice ethical responsibilities within computer systems, as well as take appropriate security measures.

The classes will be held in a computer room; therefore, it must be ensured that all cords are safely stored to remove any trip hazards. Additionally, the teacher should ensure that students practice correct posture and take regular breaks from the computers to avoid repetitive strain injuries.</td></tr>
<tr><td colspan="2">Prior Learning Experiences
Students have completed Technology Mandatory in Stage 4</td></tr>
</table>

Students have completed a graphic design unit of work in Term 1, which has provided them with basic skills and knowledge within graphic design software and its layout, as well as elements and principles. Additionally in Term 1, they have developed mini portfolios consisting of statement of intents, small amounts of research, and evaluations, which have given them skills in theoretical tasks seen within this unit of work.

Diverse and Inclusive Teaching and Learning Requirements:

Due to the nature of this unit, which spends a large portion of time using specific software, differentiation can be difficult however, wherever possible, considerations will be applied for diverse and inclusive teaching.

- The school, and the unit itself, provides an opportunity for support staff to work alongside student who require additional assistance.
- Assignment portfolios will be scaffolded to provide sufficient guidance for students.
- Extension tasks are available at the end of each 5-week period to allow high potential students the opportunity for additional learning
- Learning activities will vary between teacher-led, self-directed, independent and collaborative work.
- A range of input devices are available to support students in drawing/creating their designs
- Where necessary and applicable throughout the unit, the inherently ICT nature of this unit allows for individual adjustments to be made in line with NESA requirements for students with disability requirements and/or provisions.
- Assessment Task 2.2 will have slightly less scaffolding than Task 2.1, in an attempt to encourage all students to develop their own portfolio building skills, which are required in the HSC.

Resources:

- Assessment Task 2.1 Portfolio Scaffold
- Assessment Task 2.2 Portfolio Scaffold
- Weekly worksheets (Excluding weeks 8-10)
- YouTube Videos:
  - **The History of Animation – Types of Animation Styles explained:** https://www.youtube.com/watch?v=LmyRZR8MaJI
  - **12 Principles of Animation video:** https://www.youtube.com/watch?v=uDqjIdI4bF4
  - **Sculpting with Aluminium Foil: Weta Workshop's Kim Beaton:** https://www.youtube.com/watch?v=2elXD7ajmlA
- Hardware: Computers, Projector, Microphones, Graphics Tablet, Mouse, Keyboard, Camera
- Software: Adobe Audition (Audio), Adobe Illustrator (2D Drawing), Adobe Animate (2D Animation), Blender (3D Animation), Microsoft word
- *Unit 2: Animation Basics* PowerPoint slides
- Storyboarding Scaffold Worksheet (Online and paper copy)
- Extension Task Scaffold: *Stop Motion Basics*
- Generative AI: ChatGPT, etc

Assessment Tasks

	Outcomes Assessed	Description	Weighting
Assessment Task 2	**P2.1** Describes and uses safe working practices and correct workshop equipment maintenance techniques **P3.1** Sketches, produces and interprets drawings in the production of projects **P3.2** Applies research and problem-solving skills **P3.3** Demonstrates appropriate design principles in the production of projects **P4.1** Demonstrates a range of practical skills in the production of projects **P4.2** Demonstrates competency in using relevant equipment, machinery and processes **P5.2** Uses appropriate documentation techniques related to the management of projects	**This unit of work will be assessed by Assessment Task 2, which is broken down into two smaller tasks. Task 2.1 will be a 2D animation-based task with a portfolio and Task 2.2 will be a 3D based task with a portfolio.**	**20%**
2.1: 2D Animation Mini Video + Portfolio	**P2.1** Describes and uses safe working practices and correct workshop equipment maintenance techniques **P3.3** Demonstrates appropriate design principles in the production of projects **P4.1** Demonstrates a range of practical skills in the production of projects **P4.2** Demonstrates competency in using relevant equipment, machinery and processes **P5.2** Uses appropriate documentation techniques related to the management of projects	Assessment Task 2.1 will consist of a very basic 2D animation of the student's choosing and a mini portfolio. The 2D animation will be developed during class time following teacher guidance and will be required to demonstrate the skills that students have developed over the 5-week period. Each week, they will build their skills by continuing to develop their animation. The mini portfolio will consist of the theoretical components focused on each week. These elements	10%

Commented [AB1]: This summative assessment table outlines the contents and requirements of the summative assessments throughout the unit. In week one and six, the expectations, content and all relevant information about the respective assessment task is presented to the students.

		will be Statement of Intent, a base-level time plan, record of production, WHS Considerations and Evaluation.	
2.2: 3D Animation Mini Video + Portfolio	**P3.2** Applies research and problem-solving skills **P3.3** Demonstrates appropriate design principles in the production of projects **P4.1** Demonstrates a range of practical skills in the production of projects **P4.2** Demonstrates competency in using relevant equipment, machinery and processes **P5.2** Uses appropriate documentation techniques related to the management of projects	Assessment Task 2.2 will comprise of a very basic 3D animation, and a mini portfolio. The 3D animation will be developed during class time, following teacher guidance and will be required to demonstrate the skills that they will develop over the 5-week period. Each week, they will learn new skills to apply to their animation, with the fifth week allowing for independent skill development. The portfolio will consist of the theory focuses from each week. These elements will be Statement of Intent, Research, Storyboarding, basic prototyping and Evaluation.	10%

Week	Students learn about:	Students learn to:	Teaching, Learning and Assessment	Student Diversity
1	Appropriate software relevant to the project in the areas of: authoring • publishing • sound creation/capture/editing • image creation/capture/editing • video creation/capture/editing • text creation/capture/editing • animation creation/capture/editing • 2D/3D drawing • web page design Multimedia computer systems • processor speed • RAM • graphics cards • storage • motherboards • screen type and resolution • sound cards input devices, including: • keyboard • mouse – joystick • game controller • graphics tablet • microphone – scanners output devices: • screens • printers (ink-jet and laser) • projectors image creation/editing/conversion • bitmap • vector • scanning • formats • compression	• recognise computer hardware typically used in multimedia computer systems • identify computers and related hardware components • describe multimedia software and related memory, processing and storage requirements • understand and apply the procedures associated with the correct use of a computer system • identify and use input and output devices in conjunction with specific multimedia software • investigate and use a range of software suitable for the creation, editing and publishing of multimedia projects • identify and use planning processes related to a range of multimedia presentations • investigate and discuss the processes of obtaining, creating and modifying images, sound and text • produce and manipulate digital images • use presentation techniques and strategies in multimedia • apply principles of design in the planning and production of multimedia presentations	**Summary:** T: Introduces Ss to 2D animation basics, software and hardware. Advises Ss of unit expectations, assessments and learning goals. **Resources:** • Week 1 Worksheet • Computers • Projector • Assessment Schedule • Assessment Task 2.1 Portfolio Scaffold • 2D design software • PowerPoint Presentation • **The History of Animation – Types of Animation Styles explained:** https://www.youtube.com/watch?v=LmyRZR8MaJI **Teaching and Learning Activities:** ➢ Teacher to introduce and discuss expectations and learning goals of the new unit so that students have a clear understanding of their requirements. ➢ Introduction to the basics of animation: Students watch. **The History of Animation – Types of Animation Styles explained** ➢ Students to fill in the table and open-ended questions in the week 1 worksheet, using the video and PowerPoint presentation information, with additional research if required. ➢ Teacher to discuss Assessment Task 2.1 requirements, providing the notification sheet and scaffold for students. ➢ Students to recap Term 1: Graphic Design and design an element of their project for Assessment Task 2.1, they may have the choice between a background or a character. ➢ Students to complete WHS section of worksheet as Teacher clearly demonstrates Work, Health and Safety regulations and expectations for this unit ➢ Students to commence an on-going evaluation diary, share it with the teacher and create an entry for week 1.	➢ Students with additional learning requirements and IEPs will be provided with varied scaffolding in worksheets and Assessment Tasks ➢ Independent work will allow teac to provide additional explanations students who require such. ➢ Students will allowed to work in pairs on the worksheet to provide extra support ➢ A combinatio aural, visual a written work supports dive learning style ➢ An on-going diary allows on-going feedback opportunities. ➢ Extension and Create questi will be provide the Week 1 worksheet for finishers

Commented [AB2]: The two assessment task scaffolds within this unit of work are highly valuable resources for effective student learning. This resource has the ability to be quickly modified to have more or less scaffolding to accommodate for differing learner needs. Additionally, this resource offers a logical structure that students are able to follow, enabling them to clearly understand the expectations and outcomes.

Commented [AB3]: At the beginning of the unit, the teacher provides students with a clear explanation of the learning goals and outcomes in the unit. This provides students with an in-depth understanding of the expectations within the unit, and the skills that they will be working towards achieving. To aid in achieving these learning goals, scaffolding within assessment tasks and classwork encourage students to align their learning directly with outcomes and aid students with learning diversity

Commented [AB4]: Formative strategies are important in this unit as they allow the teacher to monitor student progress, and provide support to struggling students, as well as aiding students to take responsibility for their own skill development and progress. On-going evaluations aid students in monitoring their work throughout the unit, ensuring the students are remaining on track. These on-going evaluations further build students' evaluative abilities and assist them in summative assessment tasks.

Week	Students learn about:	Students learn to:	Teaching, Learning and Assessment	Student Diversity
2	appropriate software relevant to the project in the areas of: authoring • publishing • sound creation/capture/editing • image creation/capture/editing • video creation/capture/editing • text creation/capture/editing • animation creation/capture/editing • 2D/3D drawing • web page design image creation/editing/conversion • bitmap • vector • scanning • formats • compression	• describe multimedia software and related memory, processing and storage requirements • understand and apply the procedures associated with the correct use of a computer system • manipulate and integrate data between a range of software applications • identify and use input and output devices in conjunction with specific multimedia software • investigate and use a range of software suitable for the creation, editing and publishing of multimedia projects • identify and use planning processes related to a range of multimedia presentations • investigate and discuss the processes of obtaining, creating and modifying images, sound and text • produce and manipulate digital images • safely use computing equipment and associated materials	**Summary:** Ss: Begin planning and creating their AT2.1: 2D animation through software processes of morphing, tweening and keyframing T: theory focus on time plans and efficient planning strategies **Resources:** • Week 2 Worksheet • Computers & Projector • Assessment Task 2.1 Portfolio Scaffold • 2D design software (Adobe Illustrator) • Generative AI: ChatGPT, etc **Teaching and Learning Activities:** ➢ Think-Pair-Share activity: Teacher introduces time planning and students think of other effective planning strategies. Students share ideas with a partner then discuss as a class. ➢ Teacher to provide basic examples for the expectation of Assessment Task 2.1, including examples surrounding the use of generative AI for idea development. Students are encouraged to experiment with Generative AI for basic idea development, and practice building from these basic ideas. They will document their idea generation process ➢ Students to commence working on their Assessment 2.1 portfolio scaffold following teacher guidance and examples to complete a statement of intent and time plan. ➢ Students to learn about 2D animation components including Keyframes, Frame rates and Tweening. They will learn about interpolations and the two types of 2D animation style: cel-based and path-based. Students will fill the PowerPoint presentation information into their week 2 worksheet. ➢ Teacher will demonstrate an example of how to use Adobe Animate, using imported graphics, keyframes, tweening and morphing. Students to commence working on their Assessment 2.1 Animation, working through the basics of keyframing, tweening and morphing. Teacher will be available to aid students independently when required. ➢ Students to create an entry for Week 2 into their on-going evaluation diary.	➢ Students with additional learning requirements and IEPs will be provided with varied scaffolding in worksheets and Assessment Tasks ➢ A combination of aural, visual and written work supports diverse learning styles. ➢ An on-going diary allows on-going feedback opportunities. ➢ Students will have the opportunity to work in small groups or pairs on their worksheets if additional sup is required. ➢ On-going teac support is pro to assist stud who are strug with new conc ➢ Students will provided the opportunity to different types input devices suited to their needs and ab

Commented [AB5]: Active learning and student-led learning are engaged as pedagogical strategies throughout the unit, encouraging students to engage critically in their learning process. This is seen through teacher demonstrations, which lead into independent work, as students follow instructions and observations from the demonstration. This structure allows for the teacher to provide support to struggling students, and offer enrichment opportunities for high potential students. It further encourages higher order thinking through creativity and evaluation.

Commented [AB6]: Providing students with the ability to use a range of input devices allows them to tailor their skill development to their personal learning style. This aids students with cognitive and physical disabilities, as they are able to use technology more suited to their processing and physical abilities to keep up with the classwork.

Week	Students learn about:	Students learn to:	Teaching, Learning and Assessment	Student Diversity
3	image creation/editing/conversion • bitmap • vector • scanning • formats • compression input devices, including: • keyboard • mouse – joystick • game controller • graphics tablet • microphone – scanners appropriate software relevant to the project in the areas of: authoring • publishing • sound creation/capture/editing • image creation/capture/editing • video creation/capture/editing • text creation/capture/editing • animation creation/capture/editing • 2D/3D drawing • web page design	• describe multimedia software and related memory, processing and storage requirements • understand and apply the procedures associated with the correct use of a computer system • investigate and use a range of software suitable for the creation, editing and publishing of multimedia projects • identify and use planning processes related to a range of multimedia presentations • use presentation techniques and strategies in multimedia • apply principles of design in the planning and production of multimedia presentations • recognise workplace health and safety procedures • safely use computing equipment and associated materials	**Summary:** Ss: Investigate the 12 principles of animation and relevant software and industry considerations for AT2.1 T: Explains and displays examples of Record of Production (RoP), before Ss begin their own RoP processes. **Resources:** • Week 3 Worksheet • Computers & Projector • Assessment Task 2.1 Portfolio Scaffold • 2D design software (Adobe Illustrator) • 12 Principles of Animation video: https://www.youtube.com/watch?v=uDqjIdI4bF4 **Teaching and Learning Activities:** ➢ Students watch the 12 Principles of Animation video. ➢ Students to fill in information from video and PowerPoint presentation in Week 3 worksheet. They will fill out a table and will have an attempt at drawing an example of each principle. ➢ Students to continue working on the animation component of Assessment Task 2.1 following teacher guidance. They will work on practicing and applying the 12 principles of animation, such as the 'Squash and Stretch' aspect of movement. ➢ Teacher to inform and assist students in taking useful screenshots of their animation work to document for Record of Production. Teacher should provide examples for students so that they are aware of expectations and lay-out requirements. ➢ Students to create an entry for week 3 in their on-going evaluation diary.	➢ Students with additional learning requirements IEPs will be provided with varied scaffol in worksheets Assessment Tasks ➢ A combination of aural, visual and written work supports diverse learning styles. ➢ An on-going diary allows on-going feedback opportunities. ➢ Students will be provided the opportunity to use different types of input devices suited to their needs and abilities ➢ Examples ass students in visualising an scaffolding th work effective

Commented [AB7]: Students investigate industry considerations and applications of the principles of design. This is a direct link to the industrial technology syllabus through the Industry Study component. These links occur throughout the unit to create a relevant learning experience.

Commented [AB8]: Throughout the 2D animation section of this unit, content is logically sequenced to build upon knowledge from the previous week and strengthen student's skills. Each theory lesson valuably contributes to the practical lesson of the week. This is seen particularly in week 4, as students learn about the 12 principles of animation and its application in real-world design in their theory lessons. They then implement their theoretical knowledge in practical classes to build their animation skills.

<table>
<tr><th>Week</th><th>Students learn about:</th><th>Students learn to:</th><th>Teaching, Learning and Assessment</th><th>Student Diversity</th></tr>
<tr>
<td>4</td>
<td>appropriate software relevant to the project in the areas of:
authoring
publishing
sound creation/capture/editing
image creation/capture/editing
video creation/capture/editing
text creation/capture/editing
animation creation/capture/editing
2D/3D drawing
web page design
image creation/editing/conversion
bitmap
vector
scanning
formats
compression</td>
<td>identify and use input and output devices in conjunction with specific multimedia software
investigate and use a range of software suitable for the creation, editing and publishing of multimedia projects
investigate and discuss the processes of obtaining, creating and modifying images, sound and text
produce and manipulate digital images
use presentation techniques and strategies in multimedia
author a multimedia product
apply principles of design in the planning and production of multimedia presentations</td>
<td>Summary:
Ss: Expand knowledge and skills of 2D animation through Bone tool and Parent layering, continuing their RoP for their portfolio.
T: Demonstrates and provides the use of different input devices for student utilisation.
Resources:
Week 4 worksheet
Computers
Projector
Assessment Task 2.1 Portfolio Scaffold
2D design software (Adobe Illustrator)
Input Devices (Mouse, Keyboard, Graphics Tablet)
Teaching and Learning Activities:
Teacher to describe the importance of record of productions to project development. Class will complete a brainstorm together to develop ideas as to why keeping production records are necessary.
Students will, in pairs or individually, research one example of record of production planning in industry, documenting their findings in the Week 4 worksheet.
Teacher will demonstrate how the bone tool and parent layering works to create depth and dimension in animations.
Students to apply this learning in continuing with Assessment Task 2.1 under teacher guidance.
Students to create an entry for week 4 in their on-going evaluation diary.</td>
<td>Students with additional learning requirements and IEPs will be provided with varied scaffolding in worksheets and Assessment Tasks
A combination of aural, visual and written work supports diverse learning styles.
An on-going diary allows feedback opportunities.
The option of paired or individual work allows students to work to their personal needs and allows for additional support as needed
Students will be provided the opportunity to use different types of input devices suited to their needs and abilities</td>
</tr>
</table>

Week	Students learn about:	Students learn to:	Teaching, Learning and Assessment	Student Diversity
5	sound creation/editing • wave • MIDI • podcasts • compression formats/codecs appropriate software relevant to the project in the areas of: • animation creation/capture/editing • 2D/3D drawing input devices, including: • keyboard • mouse – joystick • game controller • graphics tablet • microphone – scanners output devices: • screens • printers (ink-jet and laser) • projectors internal and external storage devices: • USB drives • compact disc • digital video disc • hard drives select and operate computing packages - manipulate data between applications WHS • workplace procedures • safe handling of equipment • risk identification and hazard reduction strategies	• recognise computer hardware typically used in multimedia computer systems • describe multimedia software and related memory, processing and storage requirements • manipulate and integrate data between a range of software applications • identify and use input and output devices in conjunction with specific multimedia software • identify and use a range of storage devices • investigate and use a range of software suitable for the creation, editing and publishing of multimedia projects • investigate and discuss the processes of obtaining, creating and modifying images, sound and text • recognise workplace health and safety procedures • safely use computing equipment and associated materials	**Summary:** Ss: Learn about audio manipulation and integration. Experiment with audio for use in assessment task 2.1. Complete and submit AT2.1, including final evaluation and WHS considerations. **Resources:** • Week 5 worksheet • Computers & Projector • Assessment Task 2.1 Portfolio Scaffold • 2D design software (Adobe Illustrator) • Input Devices (Microphone, Mouse, Keyboard) **Teaching and Learning Activities:** ➢ Teacher to present a PowerPoint presentation on audio manipulation. Students will complete the week 5 worksheet during this presentation, covering software, hardware, audio recording and how to incorporate audio into animations. ➢ Students will learn about calculating frame rates and file sizes within animation, video and audio files. Teacher follows gradual release of responsibility for the calculations activity ➢ on the Week 5 worksheet. ➢ Students experiment with software (e.g. Adobe Audition) and hardware (E.g. RODE microphones) to become familiar with how audio manipulation functions. ➢ Students create their own audio recording to implement into the AT2.1 animation that they developed over the unit. Teacher will assist students with any difficulties. ➢ Teacher provides a table scaffolded example of WHS considerations, and project evaluation in a guided paragraph format. Students use their on-going evaluation diary to develop a final evaluation of AT2.1. They should consider time management, planning, project quality and limitations. ➢ Extra time in week 5 should be dedicated completing Assessment Task 2.1, allowing students to communicate with teacher for assistance or feedback. ➢ Students create an entry for week 5 in their on-going evaluation diary.	➢ Students with additional learning requirements and IEPs will be provided with varied scaffolding in worksheets and Assessment Tasks ➢ A combination of aural, visual and written work supports diverse learning styles. ➢ An on-going diary allows on-going feedback opportunities. ➢ A hands-on opportunity assists kinaesthetic learners to gain a better understanding of content but also supports visual learners. Audio and written instruction will also be provided. ➢ A Stop-Motion extension task will be provided for high achieving students and early finishers.

Commented [AB9]: Students' literacy skills are developed throughout the unit as they complete the portfolios required for the summative assessments. Within their evaluations and statement of intents, they are required to use paragraph structure and subject specific language, developing their literacy skills as a result.

Week	Students learn about:	Students learn to:	Teaching, Learning and Assessment	Student Diversity
6	appropriate software relevant to the project in the areas of: authoring • animation creation/capture/editing • 2D/3D drawing Multimedia computer systems • processor speed • RAM • graphics cards • storage • motherboards • screen type and resolution • sound cards select and operate computing packages - manipulate data between applications input devices, including: • keyboard • mouse – joystick • game controller • graphics tablet • microphone – scanners output devices: • screens • printers (ink-jet and laser) • projectors internal and external storage devices: • USB drives • compact disc • digital video disc • hard drives	• recognise computer hardware typically used in multimedia computer systems • identify computers and related hardware components • describe multimedia software and related memory, processing and storage requirements • identify and use input and output devices in conjunction with specific multimedia software • investigate and use a range of software suitable for the creation, editing and publishing of multimedia projects • produce and manipulate digital images • author a multimedia product	**Summary:** T: Introduce Ss to 3D animation basics, software, and industry uses. Introduce and explain AT2.2. Ss: Begin learning basic 3D design skills **Resources:** • Computers • Projector • Assessment Task 2.2 Portfolio Scaffold • 3D design software (Blender) • Week 6 worksheet **Teaching and Learning Activities:** ➢ Teacher will introduce 3D animation, including an introduction of the software (Blender) to be used. Students will re-visit the industry applications, software, hardware and storage systems used in animation from week 1 and present this information in the week 6 worksheet. ➢ Teacher will distribute and explain Assessment Task 2.2, focusing on the 3D requirements, as well as portfolio requirements; statement of intent, research, storyboarding and evaluation. ➢ Students will develop a statement of intent for Assessment Task 2.2 during week 6, after the release of the feedback from Assessment Task 2.1. ➢ Teacher to guide students through a tutorial on the basic layout and uses of 3D software, including basic tools and functions. Students should follow along, however endeavour to create a different item to the design being demonstrated by the teacher as they can use this as Assessment Task 2.2. ➢ Students to spend the remainer of the week undertaking trial and error processes to become familiar with the application. ➢ Students to create an entry for week 6 in their on-going evaluation diary. ➢	➢ Students with additional learning requirements and IEPs will be provided with varied scaffolding in worksheets and Assessment Tasks ➢ A combination of aural, visual and written work supports diverse learning styles. ➢ An on-going diary allows on-going feedback opportunities. ➢ Revising earlier content will assist students learning by re-iterating information. ➢ Student-led skill development will provide teacher opportunity to discuss individual feedback and provide additional assistance where required.

Week	Students learn about:	Students learn to:	Teaching, Learning and Assessment	Student Diversity
7	appropriate software relevant to the project in the areas of: authoring • animation creation/capture/editing • 2D/3D drawing image creation/editing/conversion • bitmap • vector • scanning • formats • compression	• investigate and use a range of software suitable for the creation, editing and publishing of multimedia projects • produce and manipulate digital images • use presentation techniques and strategies in multimedia • author a multimedia product • apply principles of design in the planning and production of multimedia presentations	**Summary:** T: Continue 3D animation basics, introducing camera movement animation. Ss: Develop 3D animation skills and begin research to plan AT2.2 **Resources:** • Computers • Projector • Assessment Task 2.2 Portfolio Scaffold • 3D design software (Blender) • Week 7 worksheet **Teaching and Learning Activities:** ➢ Teacher will begin the week by introducing effective research techniques. The class will look at real world examples as to the critical role research plays in design. They will be introduced to the double diamond design process. ➢ In four groups, the class will investigate the 4 parts of the double diamond design process and the critical elements of each. They will share their findings, using the information to fill in the week 7 worksheet. ➢ Students will work to apply effective research methods within their portfolio for Assessment Task 2.2 ➢ Teacher will continue to guide students through 3D animation basics, creating basic characters and/or scenery from basic shapes such as spheres and cubes, as well as camera movements to create animations. ➢ Students to create an entry for week 7 in their on-going evaluation diary.	➢ Students with additional learning requirements and IEPs will be provided with varied scaffolding in worksheets and Assessment Tasks ➢ A combination of aural, visual and written work supports diverse learning styles. ➢ An on-going diary allows on-going feedback opportunities. ➢ Work varies between class, small group and individual, which allows extra support and guidance, as well as opportunities for extension where students require.

Week	Students learn about:	Students learn to:	Teaching, Learning and Assessment	Student Diversity
8	appropriate software relevant to the project in the areas of: authoring • publishing • sound creation/capture/editing • image creation/capture/editing • video creation/capture/editing • text creation/capture/editing • animation creation/capture/editing • 2D/3D drawing • web page design video and still cameras • operation • lighting • angles/composition	• investigate and use a range of software suitable for the creation, editing and publishing of multimedia projects • investigate and use a range of software tools and techniques used in the development and publishing of websites • use presentation techniques and strategies in multimedia • author a multimedia product • apply principles of design in the planning and production of multimedia presentations • identify and use input and output devices in conjunction with specific multimedia software • set up and operate basic still and video cameras for use in small media production	**Summary:** T: Demonstrate how to sculpt 3D elements to provide detail and create depth through materials. Ss: Investigate real-world material examples and implement findings into AT2.2 **Resources:** • Computers • Projector • Assessment Task 2.2 Portfolio Scaffold • 3D design software (Blender) • Cameras **Teaching and Learning Activities:** ➢ Teacher to provide examples and explanation of similar projects and their contribution to design research and planning. This is associated with the Explore/Define component of the double diamond design process. ➢ Students will apply this learning to Assessment Task 2.2, finding three similar projects to their design concept and creating a Plus/Minus/Interesting (PMI) table to demonstrate their findings. ➢ Teacher to guide students through 3D sculpting methods to introduce more detail to characters and scenery. They will learn to apply basic materials to make items more realistic and prepare for animation. ➢ Students will experiment with real cameras and lighting to gain an understanding of how shadows and lights work on materials, before applying these findings to their 3D designs. ➢ Students to create an entry for week 8 in their on-going evaluation diary.	➢ Students with additional learning requirements and IEPs will be provided with varied scaffolding in worksheets and Assessment Tasks ➢ A combination of aural, visual and written work supports diverse learning styles. ➢ An on-going diary allows on-going feedback opportunities. ➢ High Potential students will be encouraged to develop their own materials as an extension activity ➢ Student-led learning allows the teacher with opportunity to check-in with students and provide extra support where needed.

Week	Students learn about:	Students learn to:	Teaching, Learning and Assessment	Student Diversity
9	appropriate software relevant to the project in the areas of: authoring • publishing • sound creation/capture/editing • image creation/capture/editing • video creation/capture/editing • text creation/capture/editing • animation creation/capture/editing • 2D/3D drawing • web page design storyboarding ➔ types: - linear - non-linear - hierarchical - composite ➔ applications	• investigate and use a range of software suitable for the creation, editing and publishing of multimedia projects • investigate and use a range of software tools and techniques used in the development and publishing of websites • identify and use planning processes related to a range of multimedia presentations	**Summary:** T: Demonstrate how to animate characters or elements through rigging and animation techniques. Provide examples on how to prototype, storyboard and experiment for research. Ss: Implement animation techniques into AT2.2, and create prototypes and storyboards for the AT2.2 portfolio **Resources:** • Computers • Projector • Assessment Task 2.2 Portfolio Scaffold • 3D design software (Blender) • Storyboard scaffold • Sculpting with Aluminium Foil: Weta Workshop's Kim Beaton:https://www.youtube.com/watch?v=2elXD7ajmlA **Teaching and Learning Activities:** ➢ Teacher to explain the importance of prototyping and storyboarding to the design process, following the develop/test section of the double diamond design process. ➢ Students to watch Sculpting with Aluminium Foil: Weta Workshop's Kim Beaton. They will then attempt to create their main character using aluminium foil, which they will have the opportunity to photograph for their portfolio. ➢ Students complete a basic storyboard to plan the animation that they will create with the main character that they have designed over the past three weeks, or that they will design. ➢ Teacher will guide students through basic character rigging using bones and animating processes, recapping the animation basics learnt in week 2. Students will put this into practice using the character/setting that they have designed. ➢ Students to create an entry for week 9 in their on-going evaluation diary.	➢ Students with additional learning requirements and IEPs will be provided with varied scaffolding in worksheets and Assessment Tasks ➢ A combination of aural, visual and written work supports diverse learning styles. ➢ An on-going diary allows on-going feedback opportunities. ➢ A range of prototyping and storyboarding mediums will be provided, such as aluminium foil, paper, online mediums, to allow for diverse learning styles and needs ➢ A written instruction sheet will be provided alongside the foil prototyping.

Week	Students learn about:	Students learn to:	Teaching, Learning and Assessment	Student Diversity
10	appropriate software relevant to the project in the areas of: authoring • publishing • sound creation/capture/editing • image creation/capture/editing • video creation/capture/editing • text creation/capture/editing • animation creation/capture/editing • 2D/3D drawing • web page design output devices: • screens • printers (ink-jet and laser) • projectors internal and external storage devices: • USB drives • compact disc • digital video disc • hard drives	• investigate and use a range of software suitable for the creation, editing and publishing of multimedia projects. • investigate and use a range of software tools and techniques used in the development and publishing of websites. • identify and use a range of storage devices • identify and use input and output devices in conjunction with specific multimedia software • use presentation techniques and strategies in multimedia • author a multimedia product • use presentation techniques and strategies in multimedia • author a multimedia product	**Summary:** Ss: Continue developing skills in 2D and 3D animation. Complete and submit AT2.2, including a final evaluation. T: Provide students with the opportunity for assistance and feedback, as well as an extension task if necessary. **Resources:** • Computers • Projector • Assessment Task 2.2 Portfolio Scaffold • 3D design software (Blender) • Stop Motion Extension Task Scaffold **Teaching and Learning Activities:** ➢ Students will be provided with an opportunity in week 10 to continue developing the skills that they have built in weeks 6 to 9. They will be encouraged to focus on finishing the animation, as well as developing elements of their portfolio where they may have fallen behind. ➢ Teacher will guide students through writing an evaluation for Assessment Task 2.2. Students should use the feedback that they received in Assessment Task 2.1. ➢ Students to create an entry for week 10 in their on-going evaluation diary and submit the completed diary to the teacher alongside Assessment Task 2.2.	➢ Students with additional learning requirements and IEPs will be provided with varied scaffolding in worksheets and Assessment Tasks ➢ A combination of aural, visual and written work supports diverse learning styles. ➢ An on-going diary allows on-going feedback opportunities. ➢ Student-led lessons allow the teacher the opportunity to support individual students where necessary. ➢ Extension tasks will be provide high achieving students and finishers.

Commented [AB10]: Both summative assessment tasks are critical to monitoring student understanding and skill development throughout the unit. Each task is a result of the skills that students learn over the 5-week period, demonstrating how well they were able to apply their learnings practically. These tasks aid the teacher in determining the effectiveness of teaching over the unit, as well as which students may require additional support moving into the HSC course.

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